Thaddeus Harris | Materials for the Built Env. | Homework #13 | 10/30/2024

(1) (a) $MC = \frac{23.8 - 20.3}{20.3} \cdot 100\% = 17.2\%$

(b) $AC = \frac{21.8 - 20.3}{20.3} \cdot 100\% = 7.39\%$

(c) $SM = 17.2\% - 7.39\%$

$SM = 9.81\%$

$W_N = 3500kg + 0.0981 \cdot 3500kg$

$W_N = 3843.35$

(2) (a) Soil #1 is well graded, Soil #2 is gap graded, and Soil #3 is uniform. This is because soil one is a relatively straight line on the graph, soil #2 is separated into two distinct plateaus and soil #3 is mostly in the middle with there being somewhat of a plateau in the middle.

(b) Soil #1 would have approximately 72% passing, soil #2 would have approximately 71% passing and soil #3 would have approximately 63% passing.

(c) Listed on the attached file.

Additional Problems

(1) $G_B = 2.63$ $V = 29.6\%$ $W = 5 tons = 10000 lbs$

$2.63 \frac{lbs\ agg}{lbs} \cdot 67.4 \frac{lbs\ water}{ft^3} = 177.26 lbs/ft^3 \cdot \frac{27 ft^3}{1 yd^3} = 4786 lbs/yd^3$

$10000 lbs / 4786 lbs/yd^3 = 2.09 yd^3$

$V_{Total} = 2.09 yd^3 + 2.09 \cdot .296 = 2.71 yd^3$

(2) $d = 6''$ $h = 12''$ $W_D = 1.2 lb$ $W_T = 24.7 lb$ $W_{DD} = 21.8 lb$ $W_{SSD} = 22.3 lb$

$G_{Agg} = 2.66$

$W_{Agg} = (24.7 - 1.2) lb$

$W_{Agg} = 23.5 lb$

$W_{Agg-OD} = (21.8 - 1.2) lb$

$W_{Agg-OD} = 20.6 lb$

$W_{Agg-SSD} = (22.3 lb - 1.2 lb)$

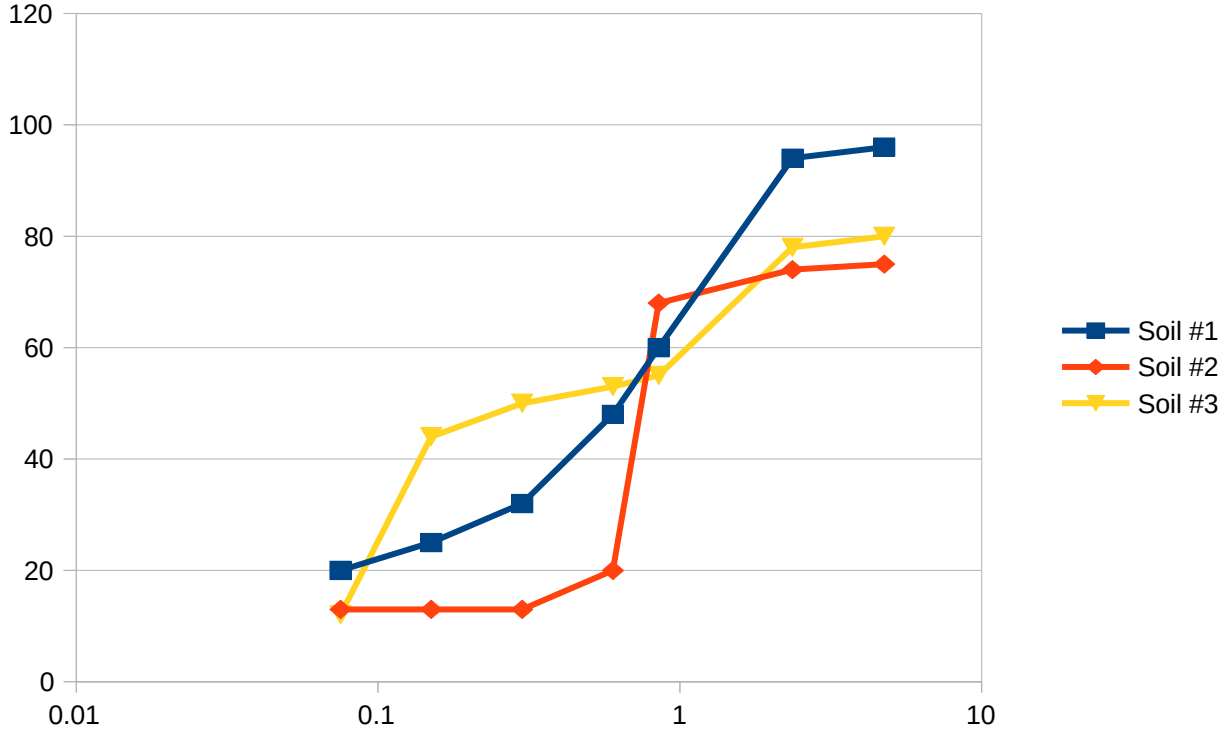
$W_{Agg-SSD} = 21.1 lb$

$MC = \frac{23.5 lb - 20.6 lb}{20.6 lb} \times 100\% = 14.1\%$

$AC = \frac{22.3 lb - 20.6 lb}{20.6 lb} \times 100\% = 8.25\%$

$SM = 14.1\% - 8.25\% = 5.85\%$

$\gamma = \frac{23.5 lb}{3^2 in \cdot \pi \cdot 12 in} = 0.0691 lb/in^3$

Sieve Size		% Passing		
ASTM	(mm)	Soil #1	Soil #2	Soil #3
No. 4	4.75	96	75	80
No. 8	2.36	94	74	78
No. 20	0.85	60	68	55
No. 30	0.6	48	20	53
No. 50	0.3	32	13	50
No. 100	0.15	25	13	44
No. 200	0.075	20	13	12

Soil #	FM
1	4.47
2	3.47
3	4.35